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|    | $\lambda = 0.2 \times 10^{-11} \text{ m}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 30 | <p><b>D</b></p> <p>Beta decay occurs when a neutron in the nucleus decays to produce a proton, a beta electron and an anti-electron-type neutrino.</p> <p>With each beta decay, the mass number of the daughter nucleus remains unchanged but the atomic number increases by one.</p> <p>Hence, after double beta decay, the mass number of the daughter nucleus remains unchanged at 130 and the atomic number increases by two to 54.</p> ${}_{52}^{130}\text{Te} \rightarrow {}_Z^A\text{Xe} + 2 {}_{-1}^0\beta$ <p>Therefore, <math>Z = 54</math> and <math>A = 130</math>.</p> |